Game-Based Optimization Method for Geo-Distributed Data Centers Under Customer Directrix Load Demand Response Mechanism

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***Abstract*—This study considers geo-distributed data centers (DCs) with diverse load profiles, electricity prices, and renewable energy availability, aggregated within their respective local virtual power plants (VPPs). To optimize demand response (DR) within such a system, a two-stage decision-making framework is proposed. First, a potential game-based method is developed to decompose the total customer directrix load (CDL) into regional sub-CDLs, aligning with grid requirements while minimizing VPP adjustment costs. Second, an evolutionary game model optimizes workload allocation strategies for DCs across spatial and temporal dimensions, considering constraints of bounded rationality and incomplete information. The proposed models are rigorously analyzed for equilibrium existence, convergence, and stability. A case study based on Google's geo-distributed DCs and local grid data demonstrates that the framework effectively enhances VPP revenues and reduces DC operational costs, especially during peak demand periods.**

***Index Terms*—Demand response, geo-distributed data centers, customer directrix load, virtual power plant, potential game, evolutionary game.**

I. Introduction

DRIVEN by the rapid expansion of digital services, alongside the growing dependence on cloud computing and big data analytics, data centers (DCs) have become integral components of modern infrastructure. Their energy demand has risen substantially, accounting for approximately 1.7%-2.2% of global electricity consumption in 2022 [1], [3], with projections indicating further growth alongside the expansion of the digital economy [2]. Transforming geo-distributed DC clusters from high energy consuming loads into flexible resources capable of beneficial interaction with the power grid through refined energy management has become one of the key scientific issues in the interdisciplinary field of power systems and information technology.

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Existing research on optimizing energy consumption in DCs focuses on two main directions. The first is dynamic voltage and frequency scaling [4], [5]. The second is energy scheduling and optimization, which aims to enhance overall energy efficiency through coordinated energy supply and workload allocation [6], [7]. Recently, integrating DCs into demand response (DR) programs has garnered significant attention [8], [9], [10], leveraging temporal and geographical variations in electricity prices and renewable energy availability. While these approaches show promise in reducing energy costs and smoothing load profiles, they face challenges in addressing the diverse needs of geo-distributed DCs, necessitating more tailored optimization frameworks.

Regarding decision-making methods for DC participation in DR, existing research has mainly adopted two solution architectures: centralized and distributed. Centralized methods model the DR scheduling of the entire DC cluster as a large-scale mathematical programming problem to seek the theoretical global optimum [23], [34], but their high computational complexity makes them difficult to scale to large-scale systems. To address the scalability challenges of centralized methods, distributed optimization algorithms have been proposed [24], [25], [33], aiming to improve solution efficiency and privacy protection through task decomposition and parallel processing. However, when constructing models, both types of solution architectures typically assume that all participants will cooperate to achieve unified system-level objectives, which significantly deviates from the real DR market where each independent entity acts as a rational individual pursuing its own profit maximization. Game theory methods, which can characterize the strategic interactions among independent entities, provide an important theoretical approach to resolve this fundamental contradiction [26].

Meanwhile, the implementation effectiveness of DR is also profoundly influenced by its underlying mechanism design. The traditional baseline-based method, as a widely adopted DR mechanism, guides behavior by compensating users for load reductions relative to a "baseline" [11]. However, such methods suffer from issues including inaccurate predictions and large errors in adjustment quantities [12], [13]. The fundamental flaw of this mechanism lies in its evaluation approach, which relies entirely on users' historical data before participating in DR. This creates arbitrage opportunities for

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rational participants to strategically increase their energy consumption levels during non-response periods to influence baseline predictions, thereby inflating baselines and obtaining improper incentives. A typical example is that energy analysis companies deliberately increase their energy consumption baseline before the grid operator's demand response (DR) event, thereby receiving fraudulent compensation based on exaggerated load reductions [30].

To overcome the aforementioned drawbacks, the customer directrix load (CDL) mechanism has been proposed as a more advanced DR paradigm [14]. Its core advantage lies in transforming the evaluation benchmark from retrospective and unreliable comparisons with historical data to tracking a publicly available target trajectory, and providing incentives based on users' tracking accuracy of this trajectory. Reference [15] constructed a large-scale DR model based on CDL to fully leverage load-side DR, improve the utilization and economics of energy storage, and verified CDL's potential in guiding diverse loads to serve system objectives. However, when applying the CDL mechanism to complex systems composed of multiple self-interested entities, two core challenges remain:

1) Adopting a unified CDL target incurs high adjustment costs due to neglecting the heterogeneity among participants [16]. Although existing research [17], [18] has begun exploring CDL decomposition methods to decompose the overall CDL target trajectory into sub-CDL target trajectories that match participants' personalized characteristics, the problem of ensuring incentive compatibility of the decomposed personalized sub-targets in competitive environments remains unresolved.
2) In reality, geo-distributed DCs are bounded rational decision-makers who cannot obtain global information and can only adjust their strategies through learning and imitation under uncertainty [26]. How to model such more realistic behavior and ensure that the system can converge to a stable state remains a research gap.

To address the aforementioned challenges, this paper constructs a two-stage game-theoretic optimization framework. For the coordination challenge of incentive compatibility, this study introduces potential game theory. By designing global constraints as penalty terms shared by all participants and embedding them into their respective utility functions, a multi-agent non-cooperative game problem is transformed into an equivalent global potential function optimization problem with the same Nash equilibrium solutions. This theoretically guarantees that when each virtual power plant (VPP) maximizes its own utility, the system state can converge to a pareto optimal equilibrium satisfying global constraints. For the behavioral challenge of bounded rationality, this study employs evolutionary game theory to characterize the process by which DCs dynamically adjust their internal resource allocation. This model does not require participants to possess complete information or global computational capabilities, but rather simulates their process of strategy learning and evolution based solely on locally observed average cost information.

In summary, the contributions of this paper are as follows:

1) To address the risks of baseline fraud and insufficient control capability in baseline-based or price-based DR, this paper proposes a method for DCs aggregated within VPPs to participate in CDL-based DR mechanisms. By enabling DCs to track target load curves to mitigate these risks, this work fills a critical gap in research on DC participation in CDL-based DR. Most importantly, this method can fully leverage the dual spatiotemporal flexibility of geo-distributed DCs to more precisely track target load curves to achieve superior DR performance.
2) To address the incentive compatibility issues that arise when multiple VPPs jointly achieve DR objectives, the multi-agent centralized optimization problem is transformed into a decentralized optimization of the potential function. By converting global constraints into shared penalty terms in each participant's utility function, the equivalence between the Nash equilibrium of decentralized decision-making and the system's Pareto optimum is ensured, thereby aligning VPPs' individual profit maximization behavior with global objective.
3) The evolutionary game model is extended to address the bounded rationality of DCs in reality. DCs can achieve sub-CDL target tracking through strategy learning based solely on local information, eliminating the need for global information or perfect rationality assumptions. Through rigorous Lyapunov stability analysis, the dynamic learning system is proven to converge stably to a unique evolutionarily stable strategy (ESS).

The paper is organized as follows: Section II introduces the system architecture. Section III details the potential game and evolutionary game models. Section IV provides equilibrium proofs. Section V presents simulation results, and Section VI concludes the study.

II. System Model and Problem Formulation

As shown in Fig. 1, the system consists of N geographic regions (set $\mathcal{N}$). Each region encompasses a local VPP_i ($i \in \mathcal{N}$), photovoltaic (PV_i), and DC_i with isomorphic computational capabilities. The independent system operator (ISO) issues the total CDL, denoted as $\tilde{p}_t$, which defines the DR target curve for each time interval $t \in \mathcal{T}$, where $\mathcal{T}$ represents the set of T discrete time periods. VPP_i is responsible for coordinating energy exchange between PV_i and DC_i within its region and participating in the DR program.

As a global load regulation target, the CDL is decomposed into regional sub-CDLs to guide load adjustments across regions. VPP_i calculates the optimal regional sub-CDL, denoted as $p_{i,t}^{\text{vpp}}$, based on the local electricity prices, renewable energy output, and supply-demand constraints, ensuring that the aggregation of all regional sub-CDLs strictly satisfies the constraint of matching the total CDL. The mathematical definitions of CDL and sub-CDL are provided in part A of this section.

In this framework, computational workloads are dynamically migrated across geo-distributed DCs through communication networks to exploit regional disparities in resource availability and operational costs, thereby achieving spatial

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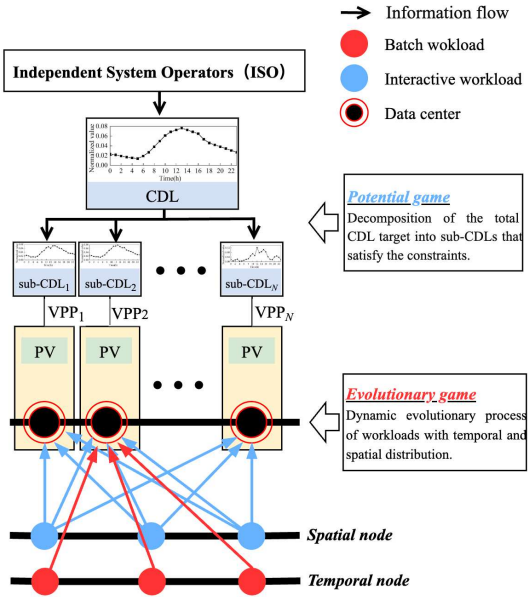


Fig. 1. Framework for geo-distributed DCs participating in demand response.

load balancing. Workloads are categorized into two distinct types: interactive workloads with stringent delay constraints, and batch workloads with flexible execution windows. Interactive workloads are preferentially scheduled to DCs that offer both low network latency and competitive operational costs. When DC_i experiences capacity constraints, elevated electricity prices, or insufficient local PV generation, its workloads are strategically redirected to alternative DC_j ($j \in \mathcal{N}$) in regions with more favorable economic conditions or abundant renewable energy resources. In contrast, batch workloads exhibit temporal flexibility analogous to traditional demand-side flexible loads, enabling their strategic scheduling across extended time horizons to flatten load profiles and maximize resource utilization efficiency. The detailed cost models governing the processing of these differentiated workload types are formulated in Part B of this section.

The overall operational process is as follows: first, the ISO issues the global CDL target, $\tilde{p}_t$, to all VPPs. Second, each VPP, acting as a rational player, uses this public target and its private local data to determine its optimal sub-CDL procurement plan $\tilde{p}_{i,t}^{\mathrm{vpp}*}$, via a potential game. Third, this personalized sub-CDL is communicated downward to the VPP's managed DC. Finally, the DC's internal system, acting as a boundedly rational agent, executes an evolutionary game. It uses the received sub-CDL as a guiding target and adjusts its server allocation strategies $x_{k,i,t}$, based on locally observed costs ($\bar{\pi}_{i,t}$) until an ESS is reached.

A. Definitions of CDL and Sub-CDL

The CDL is a per-unit signal that retains only shape characteristics, providing a load shaping target for all DR participants. It is defined as follows:

Let the load curve of a regional power grid be denoted as p_t, representing the electricity consumption during the period t.

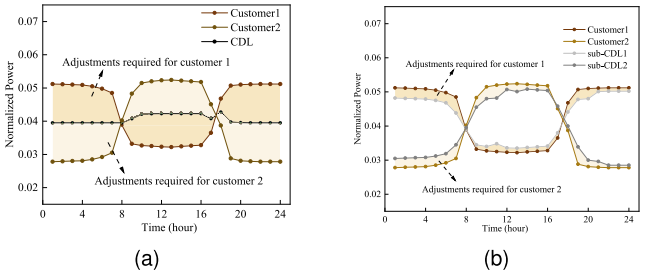


Fig. 2. Adjustment target curve comparison: (a) General method with unified CDL target; (b) Sub-CDL method with personalized targets.

Using the total electricity consumption over T periods as the base value, the normalized load curve is referred to as the load shape. The ideal load shape is defined as the CDL [14].

$$\tilde{p}_t = \frac{p_t}{\sum_{t\in\mathcal{T}} p_t} \tag{1}$$

Consider N customers participating in DR under the ISO-released CDL. Traditional approaches require each customer to adjust their load profile to match the unified CDL target, as illustrated in Fig. 2(a). This often necessitates substantial load adjustments, leading to high response costs and operational challenges.

In contrast, the proposed VPP framework decomposes the CDL into N sub-CDLs tailored to regional electricity prices, renewable generation patterns, and individual consumption characteristics. As shown in Fig. 2(b), each customer tracks their personalized sub-CDL with minimal deviation from their baseline profile, while the aggregate of all sub-CDLs still satisfies the original CDL target. This decomposition significantly reduces adjustment costs and improves participation feasibility.

The definition of the sub-CDL is similar to that in (1), but it must satisfy the condition that the sum of all sub-CDLs equals the CDL, as shown below:

$$\tilde{p}_{i,t}^{\mathrm{vpp}} = \frac{p_{i,t}^{\mathrm{vpp}}}{\sum_{i\in\mathcal{N}}\sum_{t\in\mathcal{T}} p_{i,t}^{\mathrm{vpp}}} \tag{2}$$

which satisfies

$$\sum_{i\in\mathcal{N}}\sum_{t\in\mathcal{T}} p_{i,t}^{\mathrm{vpp}} = \sum_{i\in\mathcal{N}}\sum_{t\in\mathcal{T}} \left(p_{i,t}^{\mathrm{d}} - p_{i,t}^{\mathrm{pv}}\right) \tag{3}$$

$$\sum_{i\in\mathcal{N}} \tilde{p}_{i,t}^{\mathrm{vpp}} = \tilde{p}_t \tag{4}$$

Eq. (2) defines the regional sub-CDL as a normalized share of the global CDL target, proportional to the VPP's net demand. Eq. (3) establishes the power balance constraint at the planning phase. Eq. (4) ensures that the sum of all regional sub-CDLs precisely equals the global CDL target at each time slot. Here, i is the VPP index, $p_{i,t}^{\mathrm{vpp}}$ represents the net power demand of VPP_i, $p_{i,t}^{\mathrm{d}}$ is the power demand of the data center, $p_{i,t}^{\mathrm{pv}}$ is the local photovoltaic generation, $\tilde{p}_{i,t}^{\mathrm{vpp}}$ denotes the regional sub-CDL, and $\tilde{p}_t$ is the global CDL target.

To achieve effective decomposition of the CDL, a potential game approach is adopted, mapping the local strategy choices

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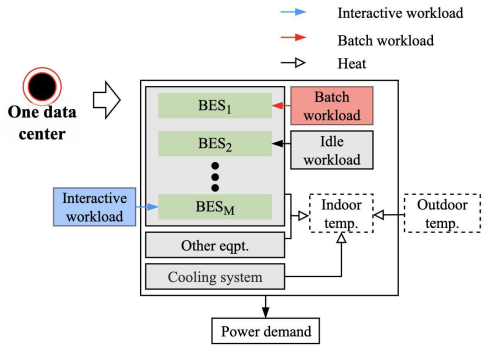


Fig. 3. Architecture of DC operation.

of each VPP in adjusting sub-CDLs to a system-wide utility optimization problem. The equilibrium solution of a potential game corresponds to the extremum of the potential function, ensuring global optimality and stability of the decomposition results. This approach avoids the pitfalls of non-uniqueness or poor convergence often seen in traditional game methods. The mathematical model of the potential game will be discussed in detail in Section III.

B. Power Consumption and Operational Cost Models of DCs

Before analyzing the workload transfer strategies of geo-distributed DCs based on evolutionary game theory, it is necessary to convert the nonlinear load model into a linear model defined by electrical variables.

As shown in Fig. 3, this study decomposes the power consumption of a single DC into three components: IT equipment power consumption, cooling system power consumption, and other power consumption.

IT equipment primarily includes servers, storage devices, and communication devices. Since back-end servers (BES) are primarily responsible for workload computation, it is essential to establish a relationship between the power consumption of BES and the processing of various workload types.

Let DC_i denote the DC managed by the i-th VPP, containing M BES units. Assume that each BES processes only one type of workload in a single time slot. The set of operational states for a BES is represented by $\mathcal{K}$. The state is denoted by $k \in \mathcal{K}$, where $k = 1$ represents processing interactive workloads, $k = 2$ represents processing batch workloads, and $k = 3$ denotes the idle state. The power consumption and cost models are as follows.

1) Power Consumption Model: The power consumption of a DC is divided into the IT equipment power consumption model, the infrastructure power consumption model, and the cooling facility power consumption model.

For IT equipment, the power consumption mainly consists of the baseline power and the additional power consumption for handling task requests. When calculating the power consumption for BESs in different operational states, the power consumption for BESs in state k is expressed in the following mixed-integer linear form:

$$p_{k,i,t}^{IT} = \begin{cases} m_{k,i,t}\left(\dfrac{n_i^E e_i^E + n_i^A e_i^A + n_i^C e_i^C}{M_i} + e_i^{\text{idle}}\right) \\ \quad + \left(\dfrac{e_i^{\text{peak}} - e_i^{\text{idle}}}{\gamma}\right)\vartheta_{k,i,t}, & k \in \{1,2\} \\ m_{k,i,t} e_i^{\text{idle}}, & k = 3 \end{cases} \tag{5}$$

$$m_{k,i,t} = \begin{cases} \left\lceil \dfrac{\vartheta_{k,i,t}}{\gamma - 1/\nu} \right\rceil, & k = 1 \\ \left\lceil \dfrac{\vartheta_{k,i,t}}{\gamma} \right\rceil, & k = 2 \\ M - \left\lceil \dfrac{\vartheta_{k,i,t}}{\gamma - 1/\nu} \right\rceil - \left\lceil \dfrac{\vartheta_{k,i,t}}{\gamma} \right\rceil, & k = 3 \end{cases} \tag{6}$$

Eq. (5) models the power consumption of IT equipment in DC_i with BESs in state k during time slot t. The first term represents baseline power consumption, while the second term accounts for additional workload-dependent power. $m_{k,i,t}$ is the number of active BESs in state k. $\vartheta_{k,i,t}$ is the processed workload volume. n_i^E, n_i^A, n_i^C are quantities of edge, aggregate, and core switches. e_i^E, e_i^A, e_i^C are their rated power consumption. e_i^{idle} and e_i^{peak} are the idle and peak power of each BES. γ is the average service rate. ν is the acceptable service delay for interactive workloads.

Eq. (6) determines the number of servers required to process different types of workloads based on M/M/1 queuing theory [20].

The power consumption of infrastructure and cooling facilities can be expressed as follows:

$$p_{k,i,t}^{BC} = e_{i,t}^{\text{inf}} + p_{i,t}^{\text{coo}} \tag{7}$$

$$\begin{aligned} p_{i,t}^{\text{coo}} &= b_1(r_{i,t}^{\text{in}} - r_{i,t-1}^{\text{in}}) \\ &= b_1\left[r_{i,t-1}^{\text{in}} e^{-\frac{\Delta t}{b_2 b_1}} + r_{i,t}^{\text{out}}\left(1 - e^{-\frac{\Delta t}{b_2 b_1}}\right)\right. \\ &\quad \left. + b_2 p_{k,i,t}^{IT}\left(1 - e^{-\frac{\Delta t}{b_2 b_1}}\right) - r_{i,t-1}^{\text{in}}\right] \end{aligned} \tag{8}$$

Eq. (7) represents the combined power consumption of infrastructure and cooling facilities. The first term $e_{i,t}^{\text{inf}}$ indicates the power consumption of infrastructure, and the second term $p_{i,t}^{\text{coo}}$ represents the power consumption of the cooling system.

Eq. (8) establishes the relationship between indoor temperature, cooling power, IT equipment power consumption, and outdoor temperature, based on the principle of energy conservation [28]. This model assumes that the DC behaves as a homogeneous thermal system with cooling resources allocated optimally to maintain appropriate indoor temperatures. b_1 is the equivalent thermal capacity coefficient of the DC. b_2 is the equivalent thermal resistance. $r_{i,t}^{\text{in}}$ denotes the internal temperature of DC_i during time slot t, while $r_{i,t}^{\text{out}}$ represents the ambient temperature outside DC_i at the same time. $r_{i,t}^{\text{in}}$ is the internal temperature of the DC_i during the time slot $t-1$. The exponential term $e^{-\frac{\Delta t}{b_2 b_1}}$ describes the natural temperature decay, representing how quickly the system approaches thermal equilibrium in the absence of additional thermal inputs.

Remark 1: Eq. (8) describes the dynamic cooling power consumption based on a first-order thermal model of the data center. The model assumes that: (i) the data center can be

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characterized by equivalent thermal capacity b_1 and thermal resistance b_2; (ii) heat transfer follows Newton's cooling law; (iii) within each time slot Δt, outdoor temperature and IT power remain constant. The equation is derived from the energy balance equation [27]:

$$b_1 \frac{dr_{i,t}^{in}}{dt} = \frac{1}{b_2}(r_{i,t}^{out} - r_{i,t}^{in}) + p_{k,i,t}^{IT}$$

The total power consumption of BESs in operational state k is expressed as:

$$p_{k,i,t} = p_{k,i,t}^{IT} + p_{k,i,t}^{BC} \tag{9}$$

$$\sum_{i \in \mathcal{N}} \vartheta_{1,i,t} = \varphi_t \tag{10}$$

where N is the total number of DC; φ_t is amount of interactive workload in time slot t.

For batch workloads, local workloads are only shifted in time and are not transferred between different DCs.

$$\vartheta_{2,i,t} = \bar{\vartheta}_{2,i,t} + \psi_{i,t} \tag{11}$$

$$\begin{cases} \sum_{t \in \mathcal{T}} \psi_{i,t} = 0 \\ \psi_{i,t} \le \psi_{\max} \end{cases} \tag{12}$$

where $\bar{\vartheta}_{k,i,t}$ is the baseline batch workload, $\psi_{i,t}$ is the transferred batch workload, $\psi_{i,t} \ge 0$ indicates workload transfer in, otherwise indicates workload transfer out, $\psi_{\max}$ represents the maximum allowable batch workload transfer within the tolerable delay.

2) DC Cost Model: The cost model for DCs comprises three principal components: workload migration costs, electricity costs, and subsidy income from DR participation.

For the migration component, we only consider interactive workload migration costs since batch workloads are processed locally. The migration component is defined as follows:

$$R_{k,i,t}^{M} = \begin{cases} \omega_1(\vartheta_{k,i,t})^2 + \omega_2(\vartheta_{k,i,t}), & \text{if } k = 1 \\ 0, & \text{else} \end{cases} \tag{13}$$

$$\frac{1}{\gamma - 1/\nu} + \zeta_i \le D_i \tag{14}$$

Eq. (13) represents the workload migration cost as a quadratic function of workload volume $\vartheta_{k,i,t}$ with coefficients ω_1 and ω_2. This convex formulation effectively models super-linear resource contention and bandwidth saturation that occur during large-scale data transfers in geo-distributed networks [26], [29]. The quality of service constraint in (14) ensures that the cumulative latency comprising both processing delay $\frac{1}{\gamma - 1/\nu}$ and network transmission delay ζ_i remains below the maximum tolerable delay D_i for workload transfer to DC$_i$.

According to (9), the electricity cost for BESs in operational state k is expressed as:

$$R_{k,i,t}^{C} = \lambda_{i,t}^{\text{vpp}} p_{k,i,t} \tag{15}$$

where $\lambda_{i,t}^{\text{vpp}}$ is the selling price of electricity for VPP$_i$ in time slot t.

The subsidy income is determined by the similarity between the DC's actual load shape and the sub-CDL established in the first stage of optimization. This similarity-based incentive mechanism bridges the planning and execution phases of our framework.

Let $p_{i,t}^{*} = \sum_{k=1}^{3} p_{k,i,t}$ denote the actual total load of DC$_i$ during time slot t. The normalized actual load shape curve, denoted as $\tilde{p}_{i,t}^{*}$, is calculated in a similar manner to (2).

To quantify how well a DC's actual load profile tracks its assigned sub-CDL target, a similarity metric is employed. This metric is based on the Euclidean distance between the two profiles and is defined as follows [14]:

$$\varepsilon_i = \begin{cases} e^{-\beta {d_i}^2}, & \varepsilon_i \ge \varepsilon_0 \\ 0, & \varepsilon_i < \varepsilon_0 \end{cases} \tag{16}$$

$$d_i = \sqrt{\sum_{t \in \mathcal{T}} (\tilde{p}_{i,t}^{*} - \tilde{p}_{i,t}^{\text{vpp}})^2} \tag{17}$$

The similarity metric in (16) uses an exponential form to provide stronger marginal incentives, thereby achieving more accurate load tracking. Here, ε_i represents the degree of similarity between the actual load shape of DC$_i$ and its target sub-CDL, with higher values indicating better matching; d_i denotes the Euclidean distance between the actual load shape and the sub-CDL, β is a sensitivity parameter controlling the decay rate of similarity with increasing distance, ε_0 is the threshold for similarity below which no incentive is provided and $\tilde{p}_{i,t}^{\text{vpp}}$ represents the sub-CDL determined in the first stage as defined in (2).

Remark 2: Two primary approaches exist for similarity calculation in CDL-based demand response: linear ($\varepsilon_i = 1 - d_i$) and exponential ($\varepsilon_i = e^{-\beta d_i^2}$). While the linear approach offers simplicity and transparency, we adopt the exponential function despite its increased complexity for the following reasons: (i) **Non-linear incentive structure**: Unlike the constant sensitivity of linear functions ($\frac{\partial \varepsilon_i}{\partial d_i} = -1$), the exponential function provides variable sensitivity ($\frac{\partial \varepsilon_i}{\partial d_i} = -2\beta d_i e^{-\beta d_i^2}$), creating stronger marginal incentives as data centers approach the target profile; (ii) **Natural mathematical properties**: The exponential function is naturally bounded to $(0, 1]$ for all $d_i \ge 0$ without requiring additional constraints, and maintains continuous differentiability essential for gradient-based optimization; (iii) **Operational flexibility**: The parameter β allows system operators to dynamically adjust sensitivity based on grid conditions, whereas linear functions offer no such tunability. These advantages justify the additional complexity, particularly for multi-agent optimization in geo-distributed systems [14], [32].

Given a base incentive price $\pi_{i,t}$, the subsidy income obtained for BESs in operational state k can be expressed as:

$$R_{k,i,t}^{S} = \varepsilon_i \pi_{i,t} p_{k,i,t} \tag{18}$$

III. Game Decision Methods and Models

This section consists of two main parts:

1) Formulating the sub-CDL decision-making problem for geo-distributed VPPs using potential game theory. By incorporating the adjustable capacities of DCs and PV facilities within each regional VPP, this approach ensures that the sum of the decomposed sub-CDLs

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matches the total grid demand, maximizing collective benefits and enhancing renewable energy utilization.

2) Developing a multi-population, multi-strategy evolutionary game model to optimize task allocation in geo-distributed DCs. Leveraging the established sub-CDLs, this model captures the propagation and evolution of task allocation strategies across different groups, enabling coordinated and efficient load scheduling over both temporal and spatial dimensions.

A. Potential Game-Based Sub-CDL Decision-Making Method

To address the decentralized coordination challenge among multiple VPPs in DR, this work formulates the sub-CDL decomposition problem as a potential game. This approach is designed to align the individual economic incentives of each VPP with the system-level objective of tracking a global CDL target, thereby ensuring incentive compatibility and convergence to a socially optimal outcome.

The set of players is $\mathcal{N} = \{\text{VPP}_1, \text{VPP}_2, \ldots, \text{VPP}_\text{N}\}$, where each player's strategy is the shape of its sub-CDL, denoted as $\tilde{p}_{i,t}^{\text{vpp}}$. The objective of each VPP is to maximize its profit, formulated as follows:

$$u_i = R_i - F_i - L \tag{19}$$

$$R_i = \lambda_{i,t}^{\text{vpp}} \left(\tilde{p}_{i,t}^{\text{vpp}} \sum_{i \in \mathcal{N}} \sum_{t \in \mathcal{T}} p_{i,t}^{\text{vpp}} + p_{i,t}^{\text{pv}} \right) \tag{20}$$

$$F_i = \lambda_{i,t}^{\text{grid}} \tilde{p}_{i,t}^{\text{vpp}} \left(\sum_{i \in \mathcal{N}} \sum_{t \in \mathcal{T}} p_{i,t}^{\text{vpp}} \right) + \lambda_{i,t}^{\text{pv}} p_{i,t}^{\text{pv}} \tag{21}$$

$$L = \sigma \sum_{t \in \mathcal{T}} \left(\sum_{i \in \mathcal{N}} \tilde{p}_{i,t}^{\text{vpp}} - \tilde{p}_t \right) \tag{22}$$

In (19), R_i represents the revenue function of the VPP from selling electricity to the DC, F_i denotes the power cost function of the VPP, and L is the penalty function. In (19)-(22), $\lambda_{i,t}^{\text{vpp}}$ is the selling price of electricity for the VPP, $\lambda_{i,t}^{\text{grid}}$ denotes the electricity price in the region where VPP_i is located during time slot t, and $\lambda_{i,t}^{\text{pv}}$ represents the selling price of PV electricity. $p_{i,t}^{\text{pv}}$ refers to the maximum power generation capacity of the PV managed by VPP_i in time slot t, and σ is the penalty factor. By incorporating the penalty function L, the original constraint (4) is internalized within each VPP's utility function, effectively transforming the constraint into a part of the optimization objective. This method allows the potential game framework to address the coordination among VPPs while ensuring that the sum of the sub-CDLs aligns closely with the total CDL provided by the ISO.

The mathematical relationship between $p_{i,t}^{\text{vpp}}$ and $\tilde{p}_{i,t}^{\text{vpp}}$ is reflected in (2).

$$\Phi = \sum_{i \in \mathcal{N}} (R_i - F_i) - L \tag{23}$$

The sub-CDL decision-making model for VPP members, defined by the potential function Φ, exhibits the characteristics of a pure potential game and represents an optimization problem with the finite improvement property. This enables the complex task of finding a Nash equilibrium to be converted into an optimization problem of determining the optimal solution of the potential function. The detailed proof is provided in Section IV.

B. Evolutionary Game-Based Decision-Making Method for DC Workload Shifting

To model the adaptive workload scheduling behavior of boundedly rational DCs, this section develops an evolutionary game framework. Recognizing that classical evolutionary game models are often limited to simpler scenarios, we extend this theory to a multi-population, multi-strategy context suitable for complex, geo-distributed systems.

In our model, the BES across the N DCs are partitioned into N distinct populations, corresponding to their geographic locations. Each population $i \in \mathcal{N}$ comprises M_i homogeneous BES units, where each unit operates in one of three states: processing interactive workloads ($k = 1$), processing batch workloads ($k = 2$), or remaining idle ($k = 3$). The proportions of BES units in each operational state are defined as follows:

$$x_{k,i,t} = \begin{cases} \dfrac{m_{k,i,t}}{M_i}, & k \in \{1, 2\} \\ 1 - \displaystyle\sum_{k=1}^{2} x_{k,i,t}, & k = 3 \end{cases} \tag{24}$$

$$\begin{cases} 0 \le x_{k,i,t} \le 1 \\ \displaystyle\sum_{k \in \mathcal{K}} x_{k,i,t} = 1 \end{cases} \tag{25}$$

By integrating (5), (7), (8), (9) and (24), let $a_1 = \frac{n_i^E e_i^E + n_i^A e_i^A + n_i^C e_i^C}{M_i} + e_i^{\text{idle}}$, $a_2 = \frac{e_i^{\text{peak}} - e_i^{\text{idle}}}{\gamma}$, $a_3 = 1 - e^{-\frac{\Delta t}{b_2 b_1}}$, the power consumption of BESs in operational state k is expressed as follows:

$$\begin{aligned} p_{k,i,t} = {} & a_1 M_i x_{k,i,t} + a_2 \vartheta_{k,i,t} + a_3 \left[r_{i,t}^{out} \right. \\ & \left. + b_2 \left(a_1 M_i x_{k,i,t} + a_2 \vartheta_{k,i,t} \right) - r_{i,t-1}^{in} \right] \end{aligned} \tag{26}$$

First, the overall cost distribution matrix of the DC system is constructed as follows:

$$\mathbf{\Pi}_t = \begin{bmatrix} \mathbf{\Pi}_{1,t} \\ \mathbf{\Pi}_{2,t} \\ \vdots \\ \mathbf{\Pi}_{i,t} \\ \vdots \\ \mathbf{\Pi}_{N,t} \end{bmatrix} \tag{27}$$

where $\mathbf{\Pi}_{i,t}$ is given as:

$$\mathbf{\Pi}_{i,t} = \left[\pi_{1,i,t}, \pi_{2,i,t}, \pi_{3,i,t} \right] \tag{28}$$

In (27), $\mathbf{\Pi}_t \in \mathbb{R}^{N \times 3}$ represents the global cost distribution matrix for all N DCs at time slot t, where each row corresponds to a specific DC's operational cost profile. The matrix captures the heterogeneous cost structures across geo-distributed DCs, reflecting variations in regional electricity prices, renewable energy availability, and workload migration costs. In (28), $\mathbf{\Pi}_{i,t}$ denotes the cost vector for DC_i at time t, where $\pi_{k,i,t}$ represents the unit operational cost unit operational cost per BES in DC_i operating in state k during time slot t. Specifically, $\pi_{1,i,t}$ denotes

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the cost of processing interactive workloads, $\pi_{2,i,t}$ represents the cost of processing batch workloads, and $\pi_{3,i,t}$ indicates the cost of maintaining servers in idle state. These costs incorporate electricity charges, workload migration penalties, and DR incentives as defined in (31).

When constructing the evolutionary game model, it is necessary to decouple the time-dependent formulas in (16) and (17) to ensure that the model can accurately capture the dynamic evolution of strategies over different time periods.

$$\varepsilon_{i,t} = \begin{cases} e^{-\beta d_{i,t}^2}, & \varepsilon_i \geq \varepsilon_0 \\ 0, & \varepsilon_i < \varepsilon_0 \end{cases} \tag{29}$$

$$d_{i,t} = \left| \tilde{p}_{i,t}^{*} - \tilde{p}_{i,t}^{\mathrm{vpp}} \right| \tag{30}$$

The time decoupling in (29)-(30) is essential because it makes each time slot's strategy evaluation independent, thereby ensuring the evolutionary game model operates correctly and avoiding violations of the Markovian property required by replicator dynamics.

According to (13), (15), and (18), and by replacing $\varepsilon_{i,t}$ with ε_i in (18), the cost function for BES_i in operational state k is as follows:

$$\pi_{k,i,t} = (R_{k,i,t}^{C} + R_{k,i,t}^{M} - R_{k,i,t}^{S})/m_{k,i,t} \tag{31}$$

Considering that the overall power consumption of the DC is quite substantial, while the power consumption per unit load is relatively small, the integer constraints in (6) are relaxed, resulting in the following transformation relationship:

$$\vartheta_{k,i,t} = \begin{cases} m_{k,i,t}\left(\gamma - \dfrac{1}{\nu}\right), & k = 1 \\ m_{k,i,t}\gamma, & k = 2 \\ 0, & k = 3 \end{cases} \tag{32}$$

Proof: For integer variables $m_{k,i,t}$, we use relaxation-and-correction approach to solve continuous problem for m^*, then set $\hat{m}_{k,i,t} = \lceil m_{k,i,t}^{*} \rceil$.

Assuming the objective function f is Lipschitz continuous with constant L in the relevant domain, we have:

$$|f(\hat{m}) - f(m^*)| \leq L \cdot \|\hat{m} - m^*\|_2 \tag{33}$$

Since $|\hat{m}_{k,i,t} - m_{k,i,t}^{*}| \leq 1$ for each variable, we have:

$$\|\hat{m} - m^*\|_2 \leq \|\hat{m} - m^*\|_1 \leq N \times K \times T = 432 \tag{34}$$

where $N = 6$ represents the number of geographic regions corresponding to Google's DC locations, $K = 3$ denotes the number of operational states that each server can adopt (processing interactive tasks, processing batch workloads, or remaining idle), and $T = 24$ corresponds to the number of time slots representing a 24-hour operational period.

Therefore:

$$|f(\hat{m}) - f(m^*)| \leq 432L$$

The key insight is that 432 rounding operations in a system of 1.2×10^6 servers represents a scale disparity of approximately $1 : 2800$. Under any reasonable cost structure for large-scale DCs, this scale difference ensures that integer relaxation errors do not significantly impact optimal decision-making.■

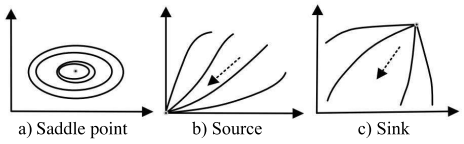


Fig. 4. Classification of internal equilibrium points.

Then, the expression for the average cost function is given as follows:

$$\bar{\pi}_{i,t} = x_{1,i,t}\pi_{1,i,t} + x_{2,i,t}\pi_{2,i,t} + \cdots + x_{K,i,t}\pi_{K,i,t} \tag{35}$$

Finally, the evolutionary dynamics of strategy selection within each population are governed by the replicator dynamics (RD) equation. Let $\dot{x}_{k,i,t}$ denote the time derivative of $x_{k,i,t}$, representing the rate of change in the proportion of BES units in DC_i operating in state k. Based on (35), the RD equation is formulated as:

$$\dot{x}_{k,i,t} = x_{k,i,t}\left(\bar{\pi}_{i,t} - \pi_{k,i,t}\right) \tag{36}$$

where $\dot{x}_{k,i,t}$ represents the proportion of BES in DC_i in operational state k. When the cost of BESs in state k falls below the average cost level, the proportion of BESs in this state will increase. The convergence conditions are as follows:

$$\left|\bar{\pi}_{i,t} - \pi_{k,i,t}\right| < \delta_i \tag{37}$$

where δ_i is the convergence precision threshold for determining when population i reaches an ESS.

Conversely, if the cost is above the average level, this proportion will decrease. This paper studies an evolutionary game model with N populations and K strategies, resulting in $N \times K$ RD equations.

Solving the RD equation can obtain the internal equilibrium point, but the internal equilibrium point is not necessarily the evolutionary equilibrium point.

Remark 3: The saddle point represents a stable critical point but does not constitute an evolutionary equilibrium. In contrast, the concentration of strategies at the source point signifies an evolutionary equilibrium, whereas the dispersion of strategies at the sink point fails to meet the criteria for evolutionary equilibrium, as illustrated in Fig.4.

C. Algorithm Flow

The two-stage game optimization algorithm we propose uses a hierarchical method, as shown in Fig. 5. Details are provided in Algorithm 1.

First, to decompose the total CDL, we construct a potential game. In this stage, regional VPPs, acting as players, solve for their optimal power procurement plans (their sub-CDL curves) by maximizing their individual utility functions u_i, which include terms for revenue, cost, and a penalty for global deviation. The Nash Equilibrium of this game yields a set of optimal, personalized sub-CDL targets $\{\tilde{p}_{i,t}^{\mathrm{vpp}*}\}$.

Secondly, to allocate workloads across the geo-distributed DCs, each DC utilizes an evolutionary game. Upon receiving its corresponding sub-CDL target, the DC finds its optimal server allocation strategy $\{x_{k,i,t}^{*}\}$. Governed by replicator

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Algorithm 1 Two-Stage Game-Theoretic Optimization Algorithm

Input: Global CDL target $\tilde{p}_t$, VPP/DC parameters.
Output: Optimal server allocations $\{x^*_{k,i,t}\}$ for all DCs.
Stage 1: Potential Game for CDL Decomposition
Initialize sub-CDL strategies $\{\tilde{p}^{\text{vpp}}_{i,t}\}$.
while Nash Equilibrium not reached **do**
    All VPPs update strategies $\{\tilde{p}^{\text{vpp}}_{i,t}\}$ by optimizing the potential function Φ based on their individual utility functions u_i.
end while
Let $\{\tilde{p}^{\text{vpp}*}_{i,t}\}$ be the resulting optimal sub-CDLs.
Stage 2: Evolutionary Game for Task Allocation
for each time slot $t \in \mathcal{T}$ **do**
    Initialize strategy proportions $\{x_{k,i,t}\}$ for all DC populations.
    while ESS not reached **do**
        For each DC i, update proportions $\{x_{k,i,t}\}$ using replicator dynamics based on current strategy costs $\pi_{k,i,t}$ and the population's average cost $\bar{\pi}_{i,t}$.
    end while
    Store the resulting ESS allocation $\{x^*_{k,i,t}\}$ for time slot t.
end for
return All optimal allocations $\{x^*_{k,i,t}\}$.

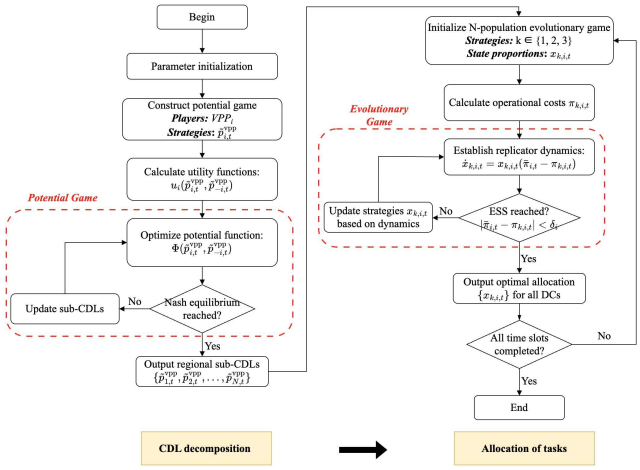


Fig. 5. CDL decomposition and workloads allocation algorithm flow.

dynamics, this process simulates the adaptive learning of a DC, where strategies evolve based on local operational costs $\pi_{k,i,t}$, until the system converges to an ESS that ensures precise tracking of the sub-CDL target established in the first stage.

D. Algorithm Complexity Analysis

The computational complexity of the proposed two-stage algorithm is determined by its sequential stages. The first stage, a potential game, converges to a Nash equilibrium with a complexity of $O(I_P \cdot N^2 T)$, where I_P denotes the number of iterations for convergence, while N and T are the numbers of VPPs and time slots, respectively. The second stage involves an evolutionary game executed for each of the T time slots, exhibiting a total complexity of $O(I_E \cdot NKT)$. Here, I_E represents the number of iterations to reach an ESS, and K is the number of strategies. Consequently, the overall complexity is given by $O(I_P N^2 T + I_E NKT)$.

IV. Proof of the Existence of Nash Equilibrium Solution

A. Proof of the Complete Potential Game Model

Potential games are a class of models that tightly couple local strategy choices with global utility optimization. In systems characterized by potential games, any individual player's strategy change influences the overall system utility. By constructing a corresponding potential function, the existence of a Nash equilibrium is guaranteed, and the challenge of finding the Nash equilibrium is transformed into the task of solving for the optimal solution of the potential function. This transformation simplifies the analysis and resolution of otherwise complex game dynamics.

The following key theorems are essential to understanding potential games:

Theorem 1: For a game $\Gamma = (N, \{Y_i\}_{i\in\mathcal{N}}, \{U_i\}_{i\in\mathcal{N}}$, if the potential function is $\Theta : G : Y \to \mathbb{R}$, then the Nash equilibrium of the game Γ corresponds to the Nash equilibrium of the potential function Θ.

Theorem 2: Every finite potential game has at least one pure strategy equilibrium.

Theorem 3: Every finite potential game satisfies the finite improvement property.

In the game $\Gamma = (N, \{Y_i\}_{i\in\mathcal{N}}, \{U_i\}_{i\in\mathcal{N}}$, where Γ represents the game model, N is the set of players, Y_i represents the strategy sets, and U_i denotes the set of utility functions, if there exists a function $G : Y \to \mathbb{R}$ that captures the changes in players' utility functions, it is called a potential function. The full potential function must satisfy the following condition:

$$U_i(y_i, y_{-i}) - U_i(y'_i, y_{-i}) = G_i(y_i, y_{-i}) - G(y'_i, y_{-i}) \tag{38}$$

In this work, the utility function for participants is constructed as shown in Eq. (19). Let $a_i = \tilde{p}^{\text{vpp}}_{i,t}$ represent the strategy of VPP_i, and a_{-i} represent the strategies of all other VPPs. When VPP_i unilaterally changes its strategy from a_i to a'_i, the change in its utility function Δu_i is:

$$\begin{aligned}\Delta u_i &= u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}) \\ &= \left[R_i(a'_i) - F_i(a'_i) - L(a'_i, a_{-i})\right] \\ &\quad - \left[R_i(a_i) - F_i(a_i) - L(a_i, a_{-i})\right] \\ &= \left[R_i(a'_i) - R_i(a_i)\right] - \left[F_i(a'_i) - F_i(a_i)\right] \\ &\quad - \left[L(a'_i, a_{-i}) - L(a_i, a_{-i})\right]\end{aligned} \tag{39}$$

Similarly, the change in the potential function $\Delta\Phi$ is:

$$\begin{aligned}\Delta\Phi &= \Phi(a'_i, a_{-i}) - \Phi(a_i, a_{-i}) \\ &= \left[\sum_{j\in\mathcal{N}} \left(R_j(a'_j) - F_j(a'_j)\right) - L(a'_i, a_{-i})\right] \\ &\quad - \left[\sum_{j\in\mathcal{N}} \left(R_j(a_j) - F_j(a_j)\right) - L(a_i, a_{-i})\right]\end{aligned}$$

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$$= \sum_{j\in\mathcal{N}} \left[\left(R_j(a'_j) - F_j(a'_j)\right) - \left(R_j(a_j) - F_j(a_j)\right)\right] - \left[L(a'_i, a_{-i}) - L(a_i, a_{-i})\right] \quad (40)$$

Therefore, the summation term in the expression for $\Delta\Phi$ simplifies significantly, as all terms for $j \neq i$ cancel out, leaving only the term for player i

$$R_j(a'_j) - F_j(a'_j) = R_j(a_j) - F_j(a_j), \quad \forall j \neq i \quad (41)$$

Substituting these into the expression for $\Delta\Phi$, we get:

$$\Delta\Phi = \left[R_i(a'_i) - R_i(a_i)\right] - \left[F_i(a'_i) - F_i(a_i)\right] - \left[L(a'_i, a_{-i}) - L(a_i, a_{-i})\right] = \Delta u_i \quad (42)$$

Consequently, the model conforms to the properties of a full potential game. Given that this is a finite game, it possesses the finite improvement property, ensuring the existence of an optimal solution to the potential function. When the potential function reaches its optimum, the system achieves Nash equilibrium.

B. Proof of the Existence of Evolutionary Equilibrium Points

To prove that the system is stable near the evolutionary equilibrium point, we introduce a Lyapunov function. The Lyapunov function for DC_i is defined as:

$$V_i(t) = \bar{\pi}_{i,t} \quad (43)$$

Because the replicator dynamics in (36) governs strategy evolution based on deviations from the population average $\bar{\pi}_{i,t}$. Thus, the average cost itself serves as a direct measure of the system's evolutionary state. At an ESS, all active strategies yield equal costs, implying a constant average cost, which is an essential characteristic for Lyapunov stability analysis.

Here, $\bar{\pi}_{i,t}$ represents the average cost of DC_i in time slot t:

$$\bar{\pi}_{i,t} = \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t} \quad (44)$$

Since the strategy proportions satisfy $0 \leq x_{k,i,t} \leq 1$ with $\sum_{k=1}^{3} x_{k,i,t} = 1$, and costs $\pi_{k,i,t} \geq 0$, we have $V_i(t) \geq 0$, confirming the Lyapunov function is positive semi-definite.

To verify stability, we compute the time derivative:

$$\dot{V}_i(t) = \frac{d\bar{\pi}_{i,t}}{dt} = \sum_{k=1}^{3} \dot{x}_{k,i,t}\pi_{k,i,t} \quad (45)$$

Substituting the replicator dynamic equation (36) into the above equation. The replicator dynamic equation is:

$$\dot{x}_{k,i,t} = x_{k,i,t}\left(\bar{\pi}_{i,t} - \pi_{k,i,t}\right) \quad (46)$$

Substituting, we obtain:

$$\dot{V}_i(t) = \sum_{k=1}^{3} x_{k,i,t}\left(\bar{\pi}_{i,t} - \pi_{k,i,t}\right)\pi_{k,i,t} = \sum_{k=1}^{3} x_{k,i,t}\bar{\pi}_{i,t}\pi_{k,i,t} - \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2 \quad (47)$$

Using the definition of the average cost in (44), we have:

$$\sum_{k=1}^{3} x_{k,i,t}\bar{\pi}_{i,t}\pi_{k,i,t} = \bar{\pi}_{i,t}\sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t} = \left(\bar{\pi}_{i,t}\right)^2 \quad (48)$$

Therefore, the time derivative of the Lyapunov function can be expressed as:

$$\dot{V}_i(t) = \left(\bar{\pi}_{i,t}\right)^2 - \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2 \quad (49)$$

Next, we introduce the variance of the costs, $\mathrm{Var}(\pi)$, defined as:

$$\mathrm{Var}(\pi) = \sum_{k=1}^{3} x_{k,i,t}\left(\pi_{k,i,t} - \bar{\pi}_{i,t}\right)^2 \quad (50)$$

Expanding the variance expression and using $\bar{\pi}_{i,t} = \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}$ and $\sum_{k=1}^{3} x_{k,i,t} = 1$, we obtain:

$$\begin{aligned}\mathrm{Var}(\pi) &= \sum_{k=1}^{3} x_{k,i,t}\left(\pi_{k,i,t}^2 - 2\pi_{k,i,t}\bar{\pi}_{i,t} + \left(\bar{\pi}_{i,t}\right)^2\right) \\ &= \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2 - 2\bar{\pi}_{i,t}^2 + \bar{\pi}_{i,t}^2 \\ &= \sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2 - \left(\bar{\pi}_{i,t}\right)^2 \end{aligned} \quad (51)$$

As a result, the relationship between $\sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2$ and $\mathrm{Var}(\pi)$ is derived as follows:

$$\sum_{k=1}^{3} x_{k,i,t}\pi_{k,i,t}^2 = \mathrm{Var}(\pi) + \left(\bar{\pi}_{i,t}\right)^2 \quad (52)$$

Substituting (52) into (49), we get:

$$\begin{aligned}\dot{V}_i(t) &= \left(\bar{\pi}_{i,t}\right)^2 - \left[\mathrm{Var}(\pi) + \left(\bar{\pi}_{i,t}\right)^2\right] \\ &= -\sum_{k=1}^{3} x_{k,i,t}\left(\pi_{k,i,t} - \bar{\pi}_{i,t}\right)^2 \end{aligned} \quad (53)$$

Since $x_{k,i,t} \geq 0$ and the squared terms are always non-negative, we have $\mathrm{Var}(\pi) \geq 0$, hence:

$$\dot{V}_i(t) \leq 0 \quad (54)$$

When $\dot{V}_i(t) < 0$, the average cost $\bar{\pi}_{i,t}$ decreases over time, and the system state evolves towards the evolutionary equilibrium point. When $\dot{V}_i(t) = 0$, it occurs only when $\pi_{k,i,t} = \bar{\pi}_{i,t}$ for all adopted strategies k, indicating that the system has reached the evolutionary equilibrium point.

In conclusion, in the evolutionary game model established in this paper, the dynamic evolution of distributed DCs' strategies will inevitably converge to the evolutionary equilibrium solution. Therefore, the dynamic system is stable, and the game can converge to the evolutionary equilibrium point.

V. Case Studies

This section demonstrates the model's implementation principles and verifies its feasibility through a practical example.

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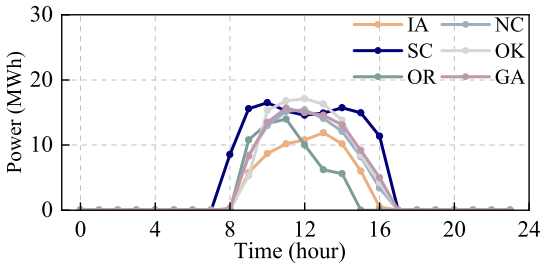


Fig. 6. PV power generation in different regions.

TABLE I
COMPREHENSIVE LIST OF KEY MODEL PARAMETERS

Category	Symbol	Value	Unit
System Configuration	N	6	-
	T	24	h
	Δt	1	h
	ε_0	0.70	-
DC Parameters	M	2×10^5	units/DC
	e^{peak}	0.20	kW
	e^{idle}	0.10	kW
	e_{it}^{inf}	2000	kW
	D_i	0.3	ms
Game Parameters	ω_1	0.8	-
	ω_2	1.5	-
	β	5	-
	σ	0.60	-

A. Experiment Settings

We consider six of Google's DCs in the United States as representative study subjects, using real electricity prices, weather, and outdoor temperature data from each location [21], [22]. The DC locations are: (1) Council Bluffs, IA, (2) Berkeley County, SC, (3) The Dalles, OR, (4) Lenoir, NC, (5) Mayes County, OK, and (6) Douglas County, GA. These locations correspond to $\text{VPP}_1 - \text{VPP}_6$ and $\text{DC}_1 - \text{DC}_6$. Photovoltaic data for each region are shown in Fig. 6. Based on recent reports on Google's server capacities, each DC's backend server capacity is set to 2×10^5 servers. Workloads are extracted from Google's cluster usage data [25], with interactive workloads managed by frontend servers centrally located in Colorado. The latency ζ_i to each region is given by the vector $[14.2, 20.4, 18.0, 20.7, 14.3, 19.8]$ [31]. For more parameter settings, refer to Table I.

To comprehensively evaluate the proposed method, we compare four DR strategies:

1) Case 1: Implements price-based DR using a decentralized approach, where each DC independently responds to real-time electricity prices without coordination.
2) Case 2: Represents baseline-based DR using a centralized approach, where the system operator calculates historical average loads as reference and determines DR compensation.
3) Case 3: Employs the CDL method with centralized target setting but distributed execution, where the system operator publishes a uniform load target for all participants to track.

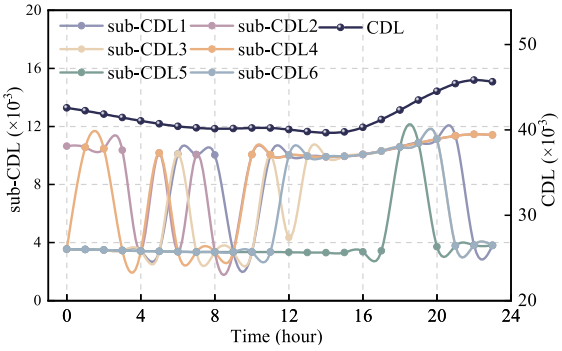


Fig. 7. The CDL decomposition results.

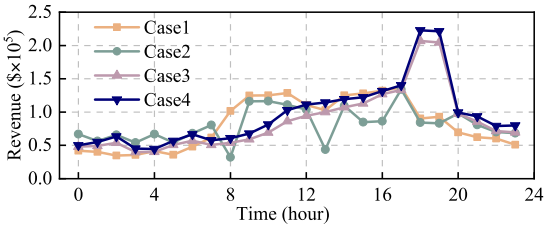


Fig. 8. Comparison of total VPP revenue.

4) Case 4: Applies our proposed sub-CDL method based on a two-stage game-theoretic framework, generating personalized regional targets that ensure incentive compatibility and system optimality.

B. Simulation Results

As shown in Fig. 7, the decomposed sub-CDLs for each region exhibit significant fluctuations. These variations reflect the model's core strategy of spatiotemporal arbitrage, where VPPs shift workloads to exploit regional differences in electricity prices and local PV generation. Although individually volatile, the sum of all sub-CDLs at each time slot strictly adheres to the total CDL target, ensuring system-level stability is maintained.

Table II compares the target DR curve tracking accuracy between the traditional CDL-based approach and our sub-CDL method. The Euclidean distance d quantifies the deviation between the actual load shape and the target curve, while the similarity ε measures the matching degree for subsidy calculation. Under the CDL-based approach, individual VPPs exhibit considerable tracking errors with Euclidean distances ranging from 0.066 to 0.120, resulting in similarity values between 0.880 and 0.934. This variation reflects the challenge faced by different regions in following a unified CDL target due to their heterogeneous characteristics, including varying electricity prices, PV generation patterns, and DC operational constraints. In contrast, the sub-CDL method achieves a significantly lower overall Euclidean distance of 0.063 and a higher similarity of 0.937. This result indicates that the personalized sub-CDLs derived through potential game optimization are better aligned with regional operational characteristics, enabling more precise load shape tracking while reducing the adjustment burden on individual DCs. The improved tracking

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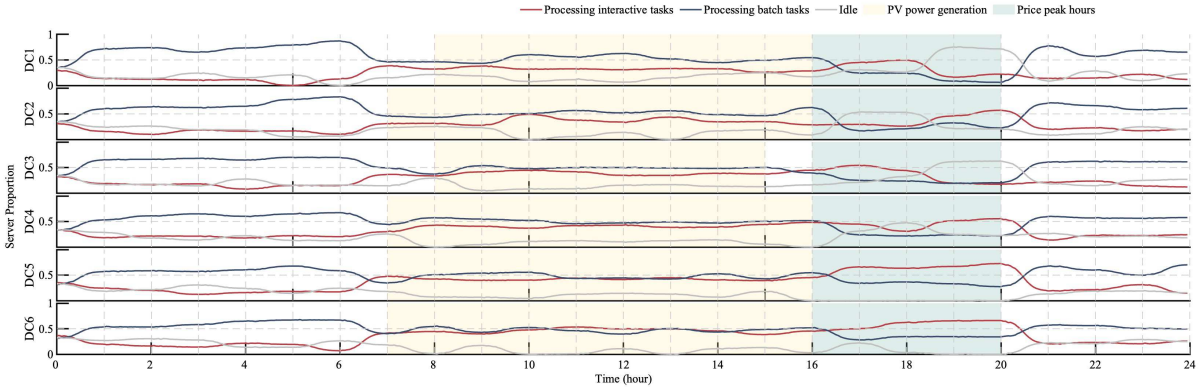


Fig. 9. Sequence of hourly ESS for server allocation in geo-distributed DCs over the dispatch period.

TABLE II

COMPARISON OF TARGET DR CURVE TRACKING ACCURACY FOR CDL AND SUB-CDL BASED METHODS

Mode of operation	**Based on CDL**						**Based on sub-CDL**
	VPP_1	VPP_2	VPP_3	VPP_4	VPP_5	VPP_6	
Euclidean distance (d)	0.066	0.120	0.113	0.107	0.105	0.118	0.063
Similarity (ε)	0.934	0.880	0.887	0.893	0.895	0.882	0.937

TABLE III

COMPARISON OF VPP'S TOTAL REVENUE ($\$ \times 10^5$)

	VPP_1	VPP_2	VPP_3	VPP_4	VPP_5	VPP_6	Total
Case1	3.53	5.61	3.31	4.14	1.80	1.97	20.36
Case2	3.34	4.90	2.13	3.52	0.87	1.46	16.22
Case3	3.89	5.61	2.84	3.85	1.00	1.53	18.72
Case4	4.07	7.68	3.81	4.77	2.07	2.43	24.83

accuracy directly translates to higher subsidy revenues and lower operational costs, validating the effectiveness of our two-stage optimization framework.

Fig. 8 compares the temporal variations of total VPP revenue under four different DR strategies over the scheduling horizon. While all strategies exhibit time-varying revenue patterns, the proposed two-stage game optimization framework (Case 4) enables VPPs to capture market opportunities more effectively, particularly achieving significantly higher revenue peaks during peak demand periods compared to alternative strategies. This superior performance is primarily attributed to the potential game-based CDL decomposition method, which allows regional VPPs to formulate optimized load profiles based on local renewable energy availability and real-time electricity prices, thereby enhancing individual VPP revenues. As shown in Table III, over the 24-hour period, the total revenue of six regional VPPs using the proposed method (Case 4) reached \$24.83 million, representing a significant 32.6% increase compared to the \$18.72 million achieved by the unified CDL method (Case 3). Furthermore, the data demonstrates that Case 4 not only improves aggregate revenue but also consistently increases the income of each individual VPP.

Fig. 9 illustrates the sequence of hourly computed ESS for the six DCs over a 24-hour dispatch period. It reveals how the equilibrium strategy adaptively adjusts to hourly variations in environmental factors, such as PV generation and electricity prices. During periods of high PV generation (time slots 8 to 16), the resulting ESS favors allocating a significant portion of server resources to batch workloads to leverage low-cost energy. Conversely, during evening hours with high

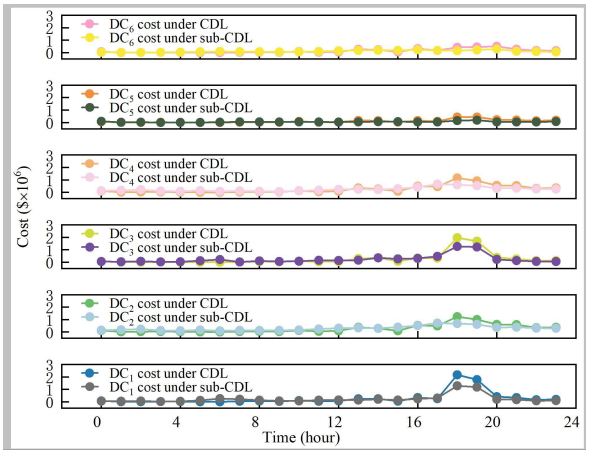


Fig. 10. Cost variations across 24 time slots for different DCs.

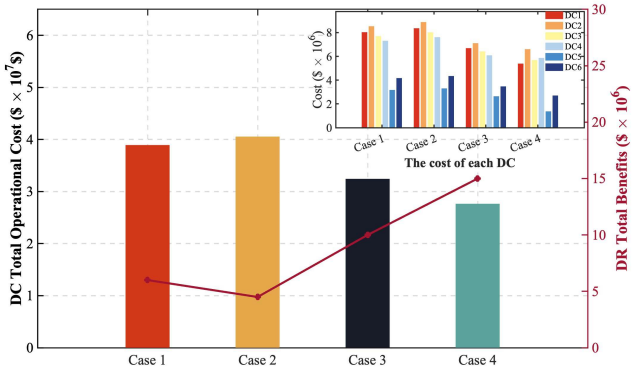


Fig. 11. Economic performance comparison of the four DR scenarios.

electricity prices (time slots 16 to 20), interactive workloads are spatially shifted to low-cost regions such as DC_5 and DC_6. Consequently, these specific DCs show a sharp decrease in idle

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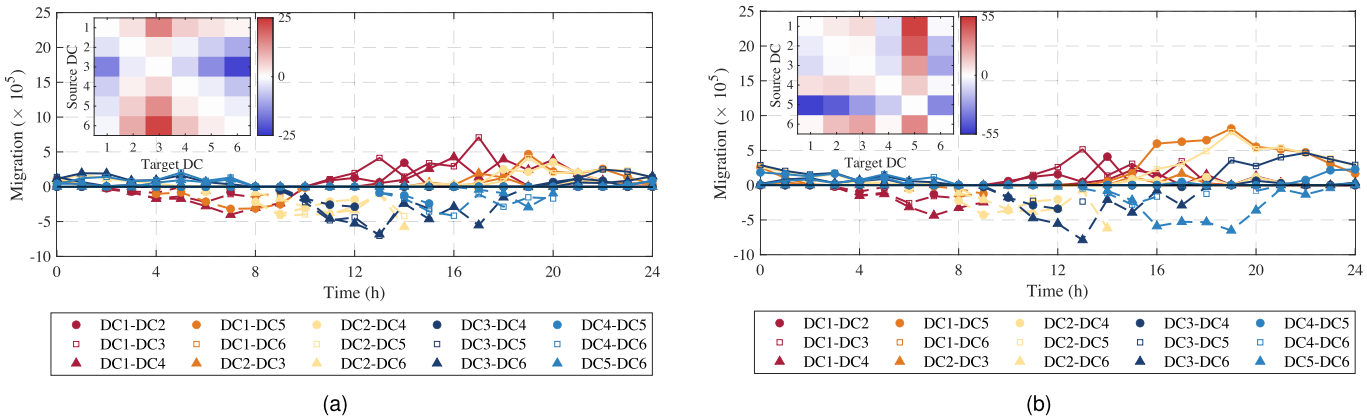


Fig. 12. Comparison of workload migration patterns under homogeneous and heterogeneous scenarios. (a) In the homogeneous scenario, migration is driven by electricity prices and PV factors. (b) In the heterogeneous scenario, migration is additionally dominated by the efficiency of DC5.

servers and an increase in the proportion of servers processing interactive workloads, effectively minimizing system-wide costs.

There are notable differences in the proportion of each strategy selected by DCs at different times of the day. Overall, during peak daytime hours, the proportion of idle state (strategy 3) increases accordingly. This variation reflects the DCs' dynamic adjustments in strategy selection in response to sub-CDL guidance and electricity price fluctuations to optimize operational costs and performance, demonstrating the effectiveness of the evolutionary game model under multi-population, multi-strategy conditions.

We compared the operational cost performance of DCs under the evolutionary game-based CDL and sub-CDL methods, as shown in Fig. 10. The operational costs for all DCs are lower under the sub-CDL method than the traditional CDL method in most time slots, particularly during peak load periods (e.g., time slots 13 to 19), where the cost differences are more pronounced. For instance, in time slot 18, the operational cost for DC_1 using the CDL method was \$2,161,800, whereas it decreases to \$1,285,600 with the sub-CDL method, a decrease of approximately 40.52%.

Fig. 10 also illustrates a downward trend in total operational costs across all DCs over the 24 time slots studied. DC 5 exhibits the largest cost reduction, reaching 48.17%, while DCs 1 and 6 see decreases of 19.48% and 22.10%, respectively. The substantial cost reductions, particularly for DC 5, result from the synergy between reduced workload migration costs, strategic exploitation of regional electricity price differentials, and enhanced DR subsidies achieved through better load-target alignment. These findings indicate that the evolutionary game strategy offers significant advantages in reducing operational costs and optimizing load distribution.

Fig. 11 presents a comprehensive economic analysis of the four DR scenarios. As illustrated, the main plot shows the total DC operational cost and corresponding DR benefits, with a detailed cost breakdown per DC in the inset. The results clearly indicate that Case 4 achieves the lowest net operational cost, which is strongly correlated with its ability to secure the highest DR benefits. This represents a total cost reduction of up to 31.8% compared to the highest cost Case 2. This synergistic result demonstrates the significant advantages of the proposed strategy in enhancing the overall economic efficiency of geo-distributed DC clusters.

To further validate the robustness and practical applicability of our framework against the over-simplified assumption of homogeneous servers, we conducted a comparative study with a heterogeneous setting. In this new scenario, we designated one data center DC_5 as a "high-efficiency" node to simulate a more realistic fleet of data centers with varying hardware capabilities. Specifically, we assumed its servers possess 15% higher processing efficiency (γ_i) and 15% lower power consumption ($e_i^{\text{peak}}, e_i^{\text{idle}}$) compared to the other five "legacy" DCs. All other simulation parameters remained unchanged.

The results of this comparative analysis are presented in Fig. 12, which starkly contrasts the workload migration patterns between the two scenarios. As shown in Fig. 12(a), workload migration in the homogeneous scenario is dynamic and bidirectional, driven primarily by time-varying factors such as regional electricity prices. The inset heatmap confirms this, showing a relatively balanced total net migration with no single DC dominating the system. In stark contrast, the heterogeneous scenario depicted in Fig. 12(b) exhibits a fundamentally different dynamic. The migration pattern is decisively dominated by the static, structural advantage of the high-efficiency DC5. A strong, persistent, and large-volume net flow of workloads towards DC5 from all other legacy DCs is observed throughout the day. This demonstrates that our proposed framework can intelligently identify and exploit server heterogeneity for superior system-wide optimization, thus confirming its robustness and practical value.

VI. Conclusion

This paper addresses the issue of geo-distributed DCs participating in DR and proposes a two-stage optimization method based on potential and evolutionary games. A model of DC power consumption is systematically established according to task type, fully accounting for the impact of different tasks on energy consumption. On this basis, potential game

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theory is first applied to decompose the CDL into sub-CDLs that meet the needs of regional VPPs. It is shown that when the potential function reaches its extremum, the system attains Nash equilibrium, satisfies global constraints, and achieves global optimization in load distribution. Second, a multi-population, multi-strategy evolutionary game model is developed, and Lyapunov function analysis is used to prove the stability and convergence of the strategy evolution process. Simulation results based on real data validate the effectiveness of the proposed method, showing significant improvements in both VPP revenue and DC operational cost reduction. This demonstrates the feasibility and practical value of the model for real-world energy management and load scheduling applications.

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